As an example, the explanation quantum mechanics gives for individual electrons, one by one, over time, building up the pattern of light and dark bands in Figure 4.4, is now clear. Each individual electron is described by its probability wave. When an electron is fired, its probability wave flows through both slits. And just as with light waves and water waves, the probability waves emanating from the two slits interfere with each other. At some points on the detector screen the two probability waves reinforce and the resulting intensity is large. At other points the waves partially cancel and the intensity is small. At still other points the peaks and troughs of the probability waves completely cancel and the resulting wave intensity is exactly zero. That is, there are points on the screen where it is very likely an electron will land, points where it is far less likely that it will land, and places where there is no chance at all that an electron will land. Over time, the electrons' landing positions are distributed according to this probability profile, and hence we get some bright, some dimmer, and some completely dark regions on the screen. Detailed analysis shows that these light and dark regions will look exactly as they do in Figure 4.4.

Einstein and Quantum Mechanics

Because of its inherently probabilistic nature, quantum mechanics differs sharply from any previous fundamental description of the universe, qualitative or quantitative. Since its inception last century, physicists have struggled to mesh this strange and unexpected framework with the common worldview; the struggle is still very much under way. The problem lies in reconciling the macroscopic experience of day-to-day life with the microscopic reality revealed by quantum mechanics. We are used to living in a world that, while admittedly subject to the vagaries of economic or political happenstance, appears stable and reliable at least as far as its physical properties are concerned. You do not worry that the atomic constituents of the air you are now breathing will suddenly disband, leaving you gasping for breath as they manifest their quantum wavelike character by rematerializing, willy-nilly, on the dark side of the moon. And you are right not to fret about this outcome, because according to quantum mechanics the probability of its happening, while not zero, is absurdly small. But what makes the probability so small?